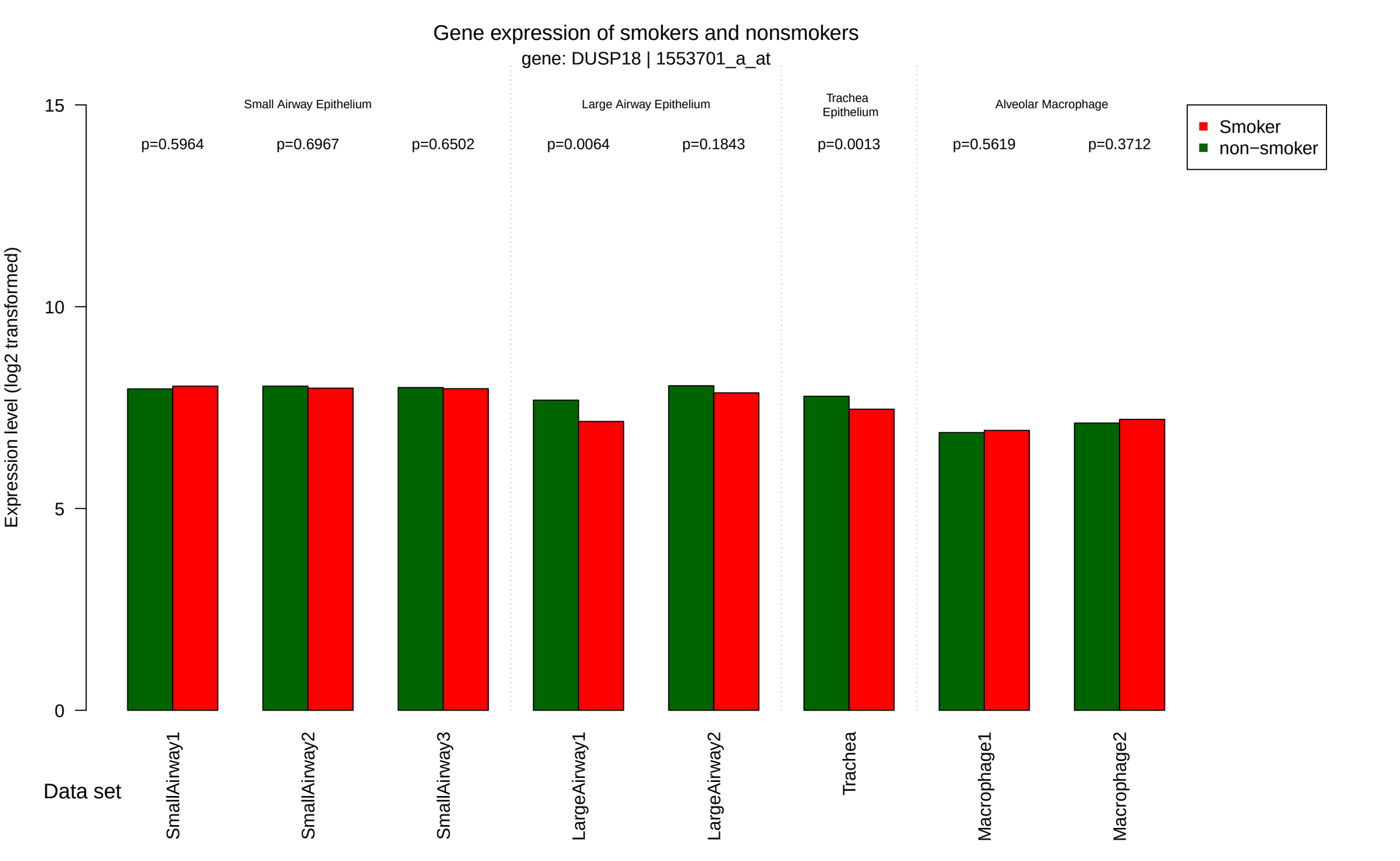
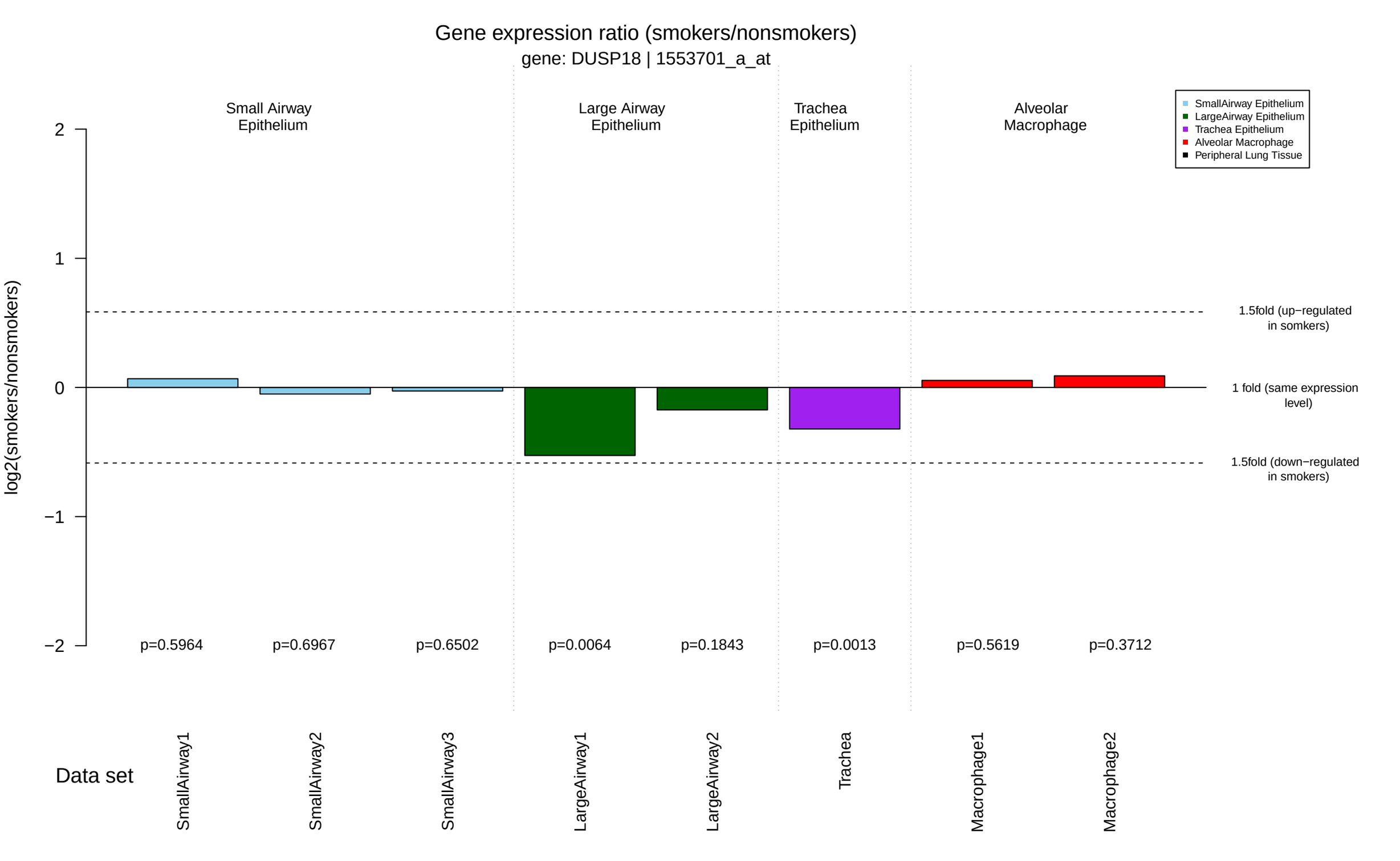
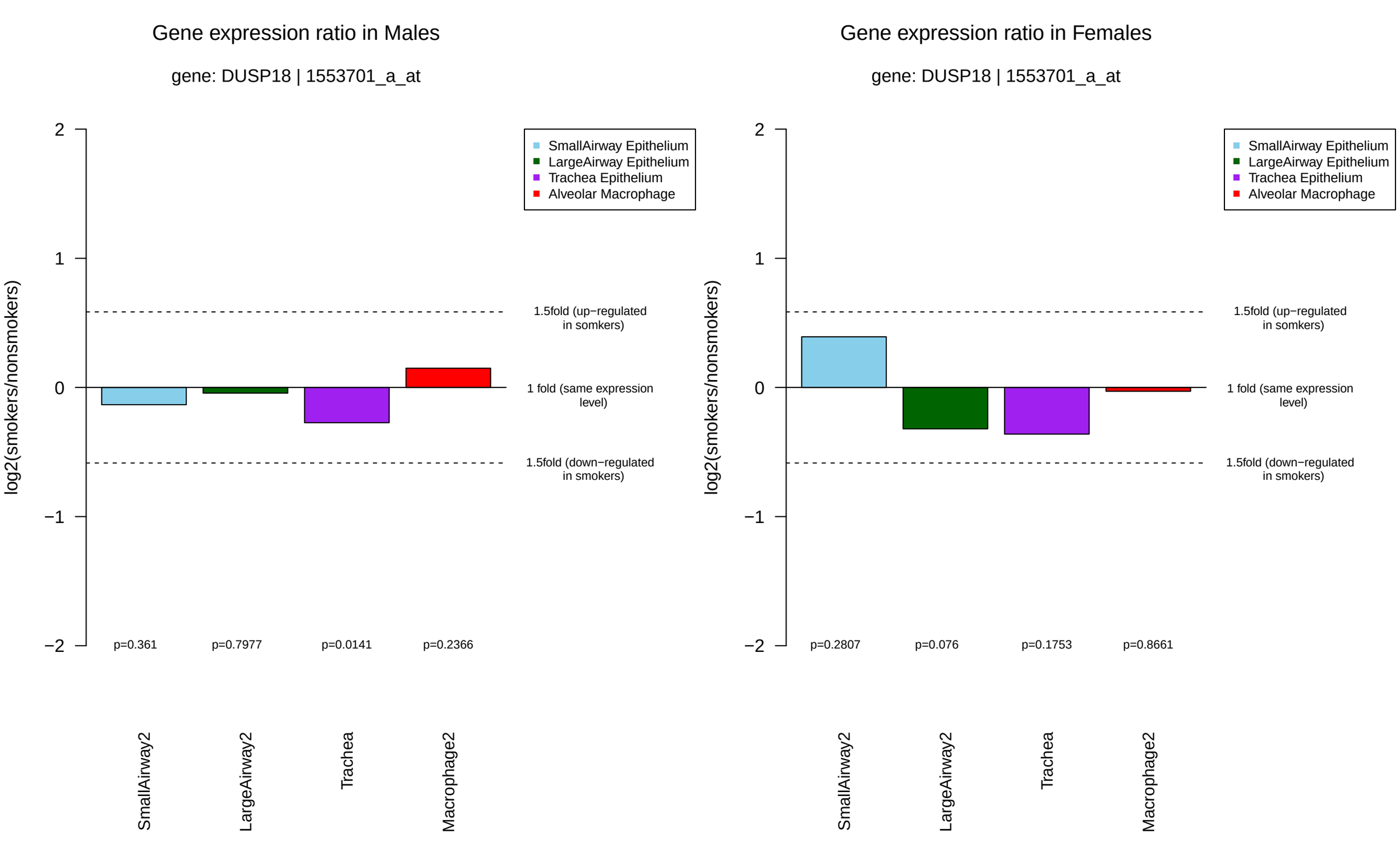
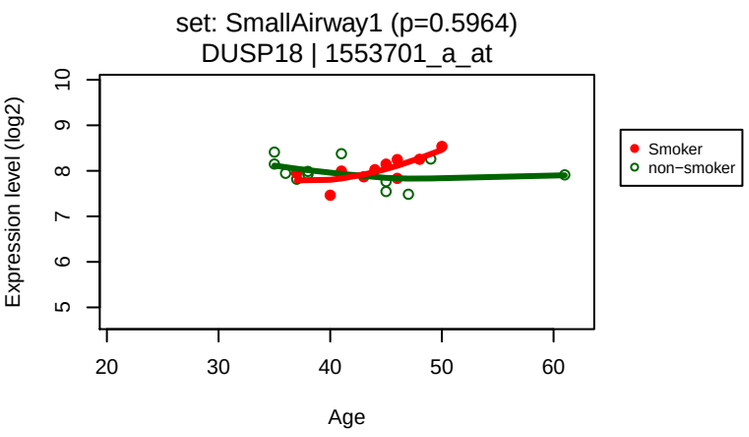
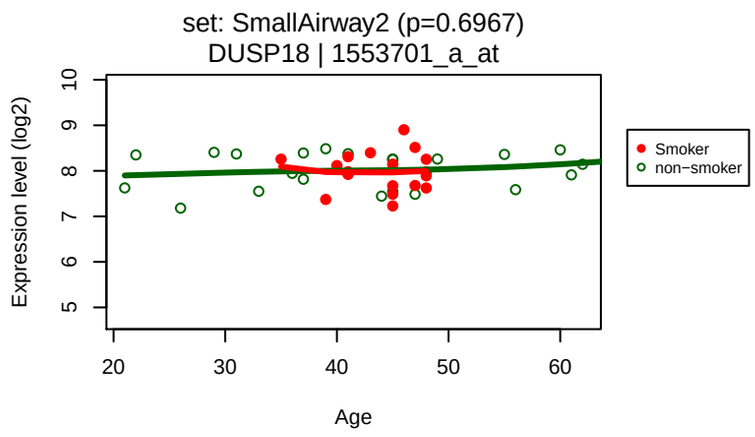
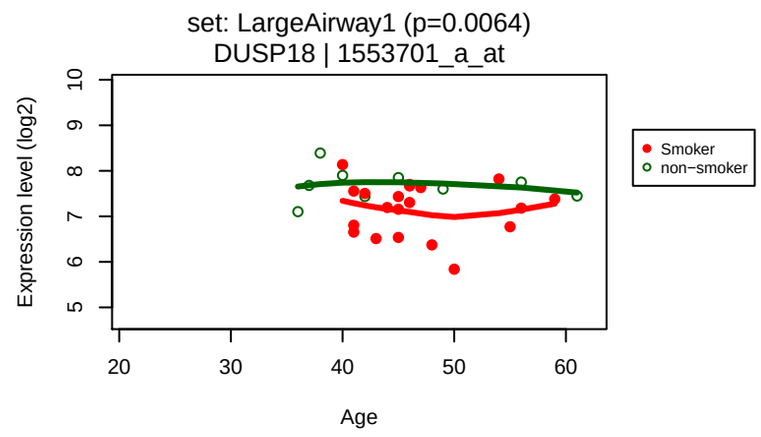

Gene expression of smokers and nonsmokers
gene: DUSP18 | 1553701_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5964
p=0.6967
p=0.6502
p=0.0064
p=0.1843
p=0.0013
p=0.5619
p=0.3712
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

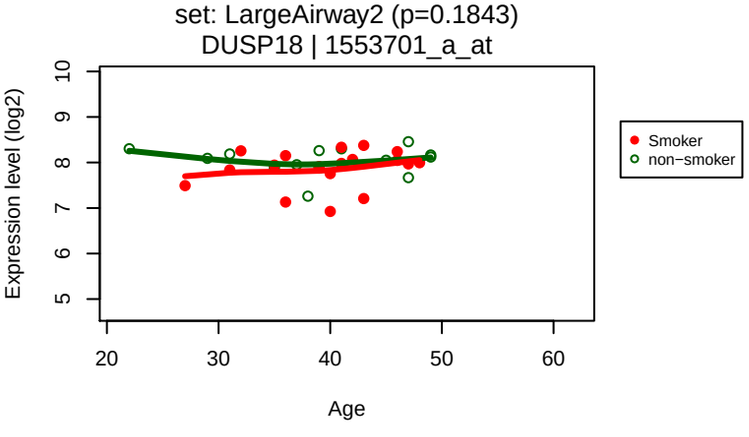

Gene expression ratio (smokers/nonsmokers)
gene: DUSP18 | 1553701_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5964
p=0.6967
p=0.6502
p=0.0064
p=0.1843
p=0.0013
p=0.5619
p=0.3712
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

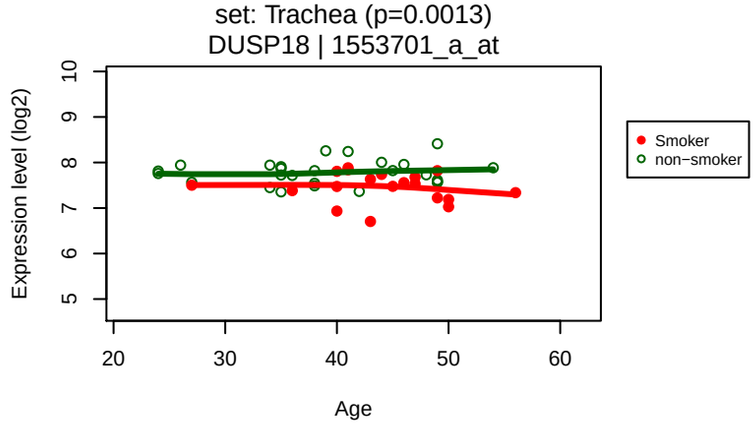

Gene expression ratio in Males
gene: DUSP18 | 1553701_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.361
p=0.7977
p=0.0141
p=0.2366
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DUSP18 | 1553701_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2807
p=0.076
p=0.1753
p=0.8661
SmallAirway2
LargeAirway2
Trachea
Macrophage2

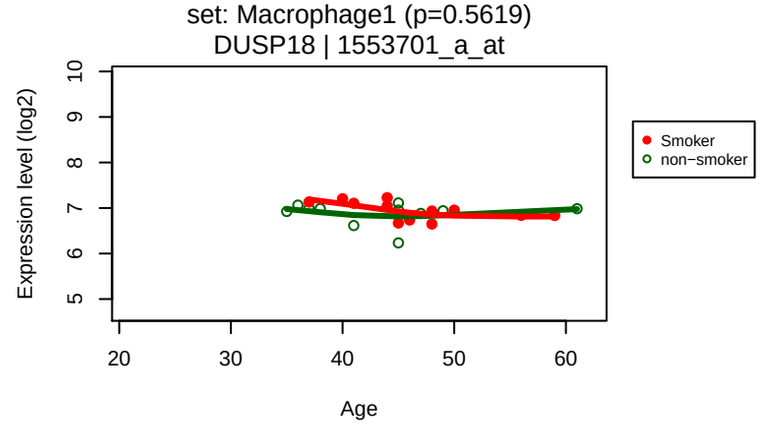

set: SmallAirway1 (p=0.5964)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker

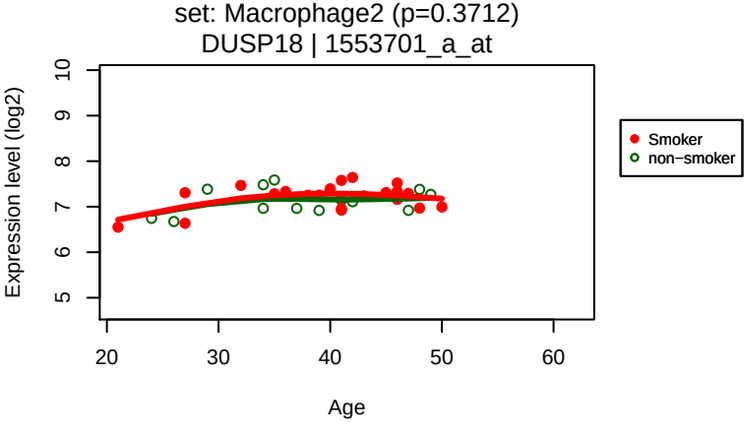

set: SmallAirway2 (p=0.6967)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker

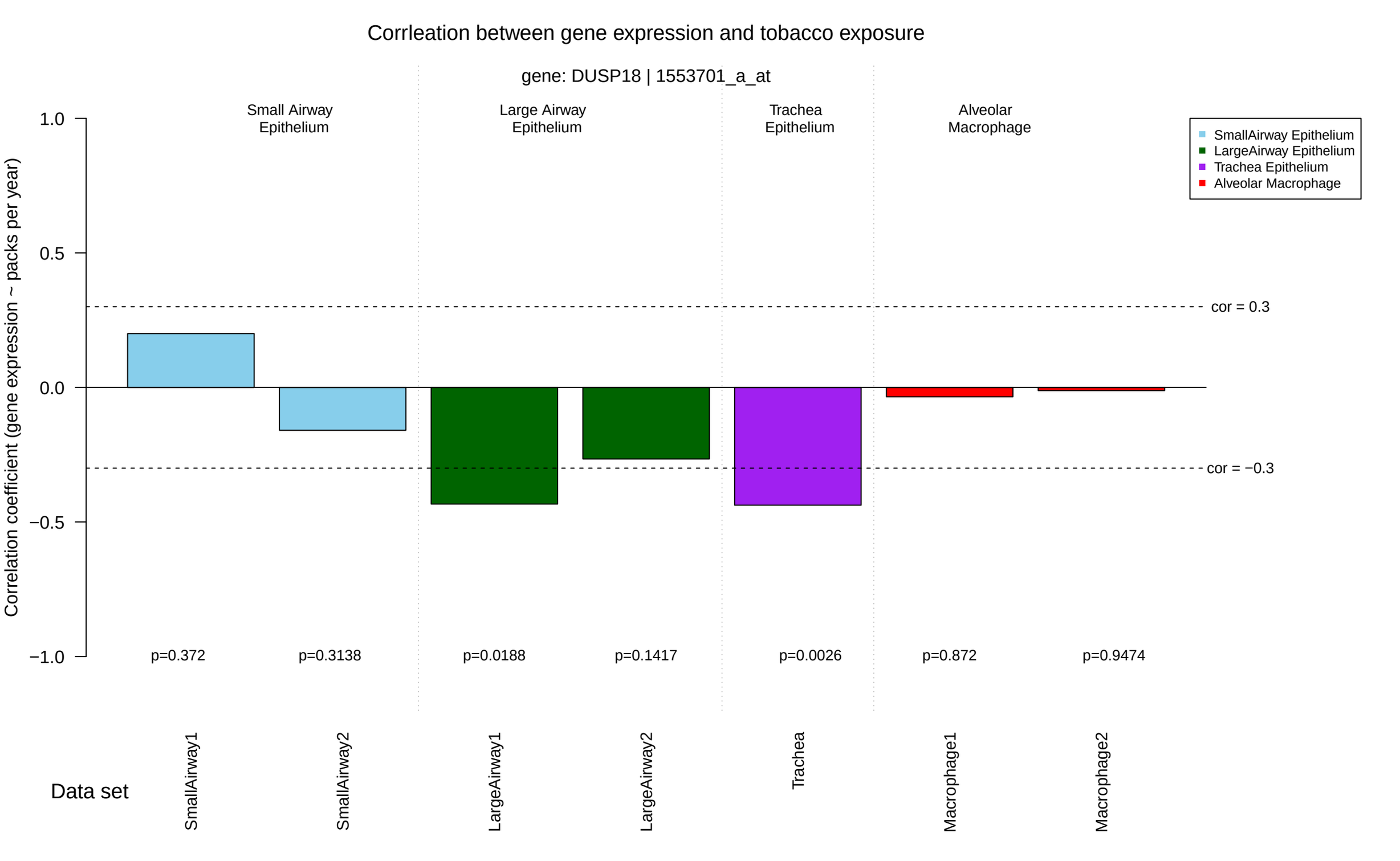

set: LargeAirway1 (p=0.0064)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1843)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5619)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3712)
DUSP18 | 1553701_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DUSP18 | 1553701_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.372
p=0.3138
p=0.0188
p=0.1417
p=0.0026
p=0.872
p=0.9474
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2